



GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Database Technologies

DS-551-MJ

Sample / Practice Question Paper

Semester II · Core · Practice Paper

Time: 3 Hours

Maximum Marks: 70

Instructions: (1) All questions carry equal weightage where indicated. (2) Figures to the right indicate full marks. (3) Draw neat diagrams / write code wherever necessary.

Q1. Answer any FIVE of the following. [5 × 2 = 10]

- a) State ACID properties.
- b) Differentiate DDL and DML.
- c) What is NoSQL?
- d) Define a document database.
- e) What is Map-Reduce?
- f) What is polyglot persistence?
- g) What is a version stamp?

Q2. Answer the following. [15]

A) Explain SQL DDL, DML and DCL commands with examples and ACID properties.
OR

B) Explain aggregate data models and distribution models in NoSQL.

Q3. Answer the following. [15]

A) Explain document databases (MongoDB) and graph databases (Neo4j) with examples.
OR

B) Explain Map-Reduce and consistency in distributed databases.

Q4. Answer the following. [15]

A) Explain schema migrations and polyglot persistence.
OR

B) Explain the factors in choosing a database technology.

